The second smooth-absolute IRMF component

Sampling error, target shift, and biased signal error after one subtraction

Draft

1 Purpose

The first-component note bounds, for the first IRMF component, the sampling error around the population target, the deterministic target shift, and their sum, the biased signal error. Those bounds use two facts that the second component loses: the summands in the score are independent, and the noise is Gaussian. The second component is fitted to the residual $Y_u - \widetilde{S}_u^{(1)}$, whose noise part is the original noise minus the first-step sampling error. That part is dependent across locations, is not Gaussian, and need not be centered.

This draft shows that the same three bounds hold at the second step once two ingredients are replaced. Bernstein's inequality is replaced by Gaussian concentration for Lipschitz functions, which needs no independence. The Gaussian closed form of the averaged loss is replaced by a Taylor bound that only uses the first two moments of the residual noise. The result is a sampling error of the same order $\sqrt{z_x/(nh_2)}$ as at step one, inflated by an explicit amplification constant, plus a carried-over remainder of order $z_x/(nh_1)$, plus a target shift that accumulates the step-one shift. Nothing contracts; the one-step rate is preserved.

Section 2 sets up the second step and the surrogate data on which the analysis runs. Section 3 states the two tools. Sections 4, 5, and 6 bound the three errors. Section 7 lists what is not proved here, including the step from two components to k. Proofs are in Appendix A. Statements of the first-component note are cited as *Note 1.*

2 The second step

Step one, recalled. Observations $Y_u = X_u + \varepsilon_u$ at n deterministic locations, with $\varepsilon_u \sim N(0, \sigma^2)$ independent, and the loss $\rho = \rho_H$ with constants $L_1 = 1$, $L_2 = c/H$, $L_3 = c/H^2$, $c = \sqrt{2/\pi}$, as in Note 1. The second step needs the first component at every observation location, so the step-one evaluation set is the design itself and the step-one union bound uses $z_x^{(1)} = x + \log(4n)$. Step one uses normalized weights $a_{v,u}^{(1)}$, the window centred at u and summing to one over v, with effective sample size N_1 and curvature lower bound λ_1 as in Note 1. Write $\widetilde{S}_u^{(1)}$, $S_u^{(1)}$, $X^{\mathrm{b}}{}_u^{(1)}$ for the observed component, the population target, and the biased signal at step one, and

$$e_u^{(1)} := \widetilde{S}_u^{(1)} - S_u^{(1)} = F_u^{-1}\nabla_u^{(1)} + R_u^{(1)}, \qquad \Delta_u^{(1)} := S_u^{(1)} - X^{\mathrm{b}}{}_u^{(1)}$$

for the step-one sampling error, its expansion, and the step-one shift. Note 1 provides an event $\Omega_1(x)$ of probability at least $1 - e^{-x}$ on which, for every u,

$$|e_u^{(1)}| \le \mathcal{E}_x^{(1)}, \qquad |R_u^{(1)}| \le \mathcal{R}_x^{(1)}, \qquad \text{and, deterministically,} \qquad |\Delta_u^{(1)}| \le \beta^{(1)}. \tag{1}$$

Step two. The second component is fitted to the residual with a smaller bandwidth $h_2 < h_1$ and normalized weights $a_{u,t}^{(2)}$, effective sample size N_2, on an evaluation set $\mathcal{T}$ with $m = |\mathcal{T}|$ points. Three fits are defined exactly as in Note 1, with the data, the clean

signal, and the averaged loss replaced by their step-two versions:

$$Y_u^{(2)} := Y_u - \widetilde{S}_u^{(1)}, \qquad \widetilde{S}_t^{(2)} := \arg\inf_z \sum_u \rho(Y_u^{(2)} - z)\, a_{u,t}^{(2)}, \tag{2}$$

$$X_u^{(2)} := X_u - X^{\mathrm{b}(1)}_u, \qquad X^{\mathrm{b}(2)}_t := \arg\inf_z \sum_u \rho(X_u^{(2)} - z)\, a_{u,t}^{(2)}. \tag{3}$$

The biased signal $X^{\mathrm{b}(2)}$ is the second output of the algorithm run on the clean signal. The population target is defined in the next paragraph, after the residual has been split into its parts.

What the residual contains. Subtracting and adding $S_u^{(1)}$ and $X^{\mathrm{b}(1)}_u$,

$$Y_u^{(2)} = \underbrace{X_u^{(2)} - \Delta_u^{(1)}}_{\bar{X}_u^{(2)}} + \underbrace{\varepsilon_u - F_u^{-1}\nabla_u^{(1)}}_{\bar{\varepsilon}_u^{(2)}} - R_u^{(1)}. \tag{4}$$

The first part $\bar{X}_u^{(2)}$ is deterministic: the clean step-two signal, displaced by the step-one shift. The second part $\bar{\varepsilon}_u^{(2)}$ is random and centered, since $\nabla_u^{(1)}$ is centered, and it is a smooth function of the Gaussian vector ε with an explicitly bounded gradient. The third part is the step-one remainder, which is small on $\Omega_1(x)$ but has no useful Lipschitz structure. The analysis therefore runs on the *surrogate residual*

$$\bar{Y}_u^{(2)} := \bar{X}_u^{(2)} + \bar{\varepsilon}_u^{(2)} = Y_u^{(2)} + R_u^{(1)}, \tag{5}$$

and the step-one remainder is handled afterwards as a data perturbation. Define the surrogate fit and the population target by

$$\bar{S}_t^{(2)} := \arg\inf_z \sum_u \rho(\bar{Y}_u^{(2)} - z)\, a_{u,t}^{(2)}, \qquad S_t^{(2)} := \arg\inf_z \sum_u \mathbb{E}\rho(\bar{Y}_u^{(2)} - z)\, a_{u,t}^{(2)}, \tag{6}$$

where the expectation is over the whole noise vector. The averaged loss now depends on u,

$$\phi_u(r) := \mathbb{E}\rho(r + \bar{\varepsilon}_u^{(2)}), \qquad \mathbb{E}\rho(\bar{Y}_u^{(2)} - z) = \phi_u(\bar{X}_u^{(2)} - z), \tag{7}$$

which Note 1 already allows. The population curvature and the centered score at the target are

$$\bar{F}_t^{(2)} := \sum_u \phi_u''(\bar{X}_u^{(2)} - S_t^{(2)})\, a_{u,t}^{(2)}, \qquad \bar{\nabla}_t^{(2)} := \sum_u \Big\{\rho'(\bar{Y}_u^{(2)} - S_t^{(2)}) - \mathbb{E}\rho'(\bar{Y}_u^{(2)} - S_t^{(2)})\Big\}\, a_{u,t}^{(2)}. \tag{8}$$

Three errors. With $\Delta_t^{(2)} := S_t^{(2)} - X^{\mathrm{b}(2)}_t$, the identity of Note 1 holds verbatim,

$$\widetilde{S}_t^{(2)} - X^{\mathrm{b}(2)}_t = \underbrace{\widetilde{S}_t^{(2)} - S_t^{(2)}}_{\text{sampling error}} + \underbrace{\Delta_t^{(2)}}_{\text{target shift}}, \tag{9}$$

and the sampling error splits further into the surrogate error and the perturbation,

$$\widetilde{S}_t^{(2)} - S_t^{(2)} = (\bar{S}_t^{(2)} - S_t^{(2)}) + (\widetilde{S}_t^{(2)} - \bar{S}_t^{(2)}). \tag{10}$$

3 Two tools

Lemma 1 (Sup-norm stability of the local fit)**.** *Let $a_u \geq 0$ sum to one and let $m(y) := \arg\inf_z \sum_u \rho_H(y_u - z)a_u$ for $y \in \mathbf{R}^n$. Then m is continuously differentiable,*

$$\frac{\partial m}{\partial y_u}(y) = \frac{\rho''_H(y_u - m(y))\, a_u}{\sum_v \rho''_H(y_v - m(y))\, a_v} \in [0,1], \qquad \sum_u \frac{\partial m}{\partial y_u}(y) = 1,$$

and for all $y, y' \in \mathbf{R}^n$,

$$|m(y') - m(y)| \leq \max_{u:\, a_u > 0} |y'_u - y_u|. \tag{11}$$

The fit is a monotone, weighted-average-like functional of the data. The lemma is used three times: to pass from the surrogate fit to the observed fit, to pass from the displaced clean signal to the clean signal, and to bound the gradient of the step-one component.

Lemma 2 (Gaussian concentration for Lipschitz functions)**.** *Let $\varepsilon \sim N(0, \sigma^2 I_n)$ and let $G : \mathbf{R}^n \to \mathbf{R}$ be differentiable with $\sup_\varepsilon \|\nabla G(\varepsilon)\|_2 \leq L$. Then for every $s > 0$,*

$$\mathbf{P}\big(|G(\varepsilon) - \mathbb{E}G(\varepsilon)| > s\big) \leq 2\exp\left(-\frac{s^2}{2\sigma^2 L^2}\right).$$

This is the Tsirelson–Ibragimov–Sudakov inequality; see for instance Boucheron, Lugosi, and Massart, *Concentration Inequalities*, Theorem 5.6. It replaces Bernstein's inequality. Independence of the summands is never used; only the Gaussian law of the original noise and a gradient bound.

4 Sampling error at step two

The step-one component as a function of the noise. Differentiating $\nabla_u^{(1)}$ with respect to ε_v gives

$$\frac{\partial \bar{\varepsilon}_u^{(2)}}{\partial \varepsilon_v} = \delta_{uv} - M_{u,v}, \qquad M_{u,v} := \frac{\rho''_H(Y_v - S_u^{(1)})\, a_{v,u}^{(1)}}{\bar{F}_u^{(1)}} \in \left[0, \frac{L_2}{\lambda_1} a_{v,u}^{(1)}\right].$$

The matrix M is the robust equivalent kernel of step one: empirical second derivatives in the numerator, population curvature in the denominator. It depends on the noise, but its entries are bounded deterministically. Let

$$\bar{a}_1 := \max_v \sum_u a_{v,u}^{(1)}, \qquad \gamma_1 := \frac{L_2}{\lambda_1}, \qquad \Gamma_1 := 1 + \gamma_1\sqrt{\bar{a}_1}. \tag{12}$$

Here $\bar{a}_1$ is the largest total weight that a single observation receives across all step-one windows; it equals one for a regular periodic design with constant window mass. Row sums of M are at most γ_1, column sums at most $\gamma_1\bar{a}_1$, so $\|M\|_{\mathrm{op}} \leq \gamma_1\sqrt{\bar{a}_1}$ and $\|I - M\|_{\mathrm{op}} \leq \Gamma_1$. Since $\lambda_1 \leq c/B \leq c/H = L_2$, always $\gamma_1 \geq 1$.

Constants. Fix a curvature lower bound $0 < \lambda_2 \leq \min_t \bar{F}_t^{(2)}$; Lemma 5 below gives an explicit choice in the small-noise regime. For $x \geq 0$ and $z_x := x + \log(4m)$, set

$$s_1^{(2)} := \sigma L_2 \Gamma_1 \sqrt{\frac{2z_x}{N_2}}, \qquad s_2^{(2)} := \sigma L_3 \Gamma_1 \sqrt{\frac{2z_x}{N_2}}, \qquad d_2 := \frac{s_2^{(2)}}{\lambda_2}. \tag{13}$$

Compared with the step-one constants of Note 1, the Bernstein linear term in αz_x is gone, the variance factor σL_2 is the Lipschitz one, and the new factor Γ_1 pays for the dependence introduced by the subtraction.

Lemma 3 (Step-two score and curvature bounds)**.** *There is an event $\Omega_2(x)$ of probability at least $1 - e^{-x}$ on which, simultaneously for every $t \in \mathcal{T}$,*

$$|\bar{\nabla}_t^{(2)}| \le s_1^{(2)}, \qquad \Big|\sum_u \rho_H''(\bar{Y}_u^{(2)} - S_t^{(2)})\, a_{u,t}^{(2)} - \bar{F}_t^{(2)}\Big| \le s_2^{(2)}. \tag{14}$$

Deterministically, the third-derivative bounds of Note 1, Lemma 4.1(ii) and (iii), hold with $C_3 = L_3/\lambda_2$.

Proposition 4 (Step-two sampling error)**.** *Suppose*

$$d_2 < 1, \qquad \frac{s_1^{(2)} L_3}{\lambda_2^2 (1 - d_2)^2} < \frac{4}{9}. \tag{15}$$

Then on $\Omega_2(x)$, simultaneously for all t, the surrogate fit is unique and

$$\begin{aligned} |\bar{S}_t^{(2)} - S_t^{(2)}| &\le \mathcal{E}_x^{(2)} := \frac{3 s_1^{(2)}}{2\lambda_2(1 - d_2)}, \\ |\bar{S}_t^{(2)} - S_t^{(2)} - (\bar{F}_t^{(2)})^{-1} \bar{\nabla}_t^{(2)}| &\le \mathcal{R}_x^{(2)} := \frac{9 L_3 (s_1^{(2)})^2}{8\lambda_2^3 (1 - d_2)^3} + \frac{s_1^{(2)} s_2^{(2)}}{\lambda_2^2 (1 - d_2)}. \end{aligned} \tag{16}$$

On $\Omega_1(x) \cap \Omega_2(x)$, which has probability at least $1 - 2e^{-x}$, also

$$|\widetilde{S}_t^{(2)} - S_t^{(2)}| \le \mathcal{E}_x^{(2)} + \mathcal{R}_x^{(1)}. \tag{17}$$

The expansion and the remainder bound are exactly those of Note 1, Proposition 4.2, applied with $C_1 = s_1^{(2)}/\sqrt{\lambda_2}$, $C_2 = s_2^{(2)}/\sqrt{\lambda_2}$, $C_3 = L_3/\lambda_2$, $D_{\min} \ge \sqrt{\lambda_2}$, and $\delta \le d_2$. The only new probabilistic input is Lemma 3. The term $\mathcal{R}_x^{(1)}$ is the price of the surrogate: the observed residual differs from the surrogate by the step-one remainder, and Lemma 1 passes that difference through the step-two fit without amplification.

5 Target shift at step two

The step-two shift has two sources. The population target averages the loss over the step-two noise $\bar{\varepsilon}^{(2)}$, which moves the minimizer as at step one. In addition, the deterministic part of the residual is $X^{(2)} - \Delta^{(1)}$ rather than $X^{(2)}$: the step-one shift has become part of the step-two signal. Introduce the clean fit of the displaced signal,

$$S_t^{(2,0)} := \arg\inf_z \sum_u \rho_H(\bar{X}_u^{(2)} - z)\, a_{u,t}^{(2)}, \qquad \Delta_t^{(2)} = \big(S_t^{(2)} - S_t^{(2,0)}\big) + \big(S_t^{(2,0)} - X^{\mathrm{b}(2)}_t\big). \tag{18}$$

The second difference is a clean fit applied to two signals that differ by at most $\beta^{(1)}$, so Lemma 1 bounds it by $\beta^{(1)}$. The first difference is the noise-averaging shift, and it needs the first two moments of $\bar{\varepsilon}^{(2)}$ and a curvature bound for the averaged loss.

Moments of the step-two noise. Since $\mathbb{E}\varepsilon_u = 0$ and $\mathbb{E}\nabla_u^{(1)} = 0$,

$$\mathbb{E}\bar{\varepsilon}_u^{(2)} = 0, \qquad \mathbb{E}(\bar{\varepsilon}_u^{(2)})^2 \le \nu^2, \qquad \nu := \sigma\left(1 + \frac{\gamma_1}{\sqrt{N_1}}\right). \tag{19}$$

The mean is exactly zero because the surrogate drops the step-one remainder; the observed residual noise $\bar{\varepsilon}^{(2)} - R^{(1)}$ has mean of order $\mathcal{R}_x^{(1)}$. That bias is accounted for in (17), not here.

Curvature of the averaged loss. Let $A_t^{(2)}$ be the width of the active range of $\bar{X}^{(2)}$ in the step-two window at t, and let $L_4 := c/H^3$, which bounds $|\rho_H'''|$. For $r > 0$, put

$$\kappa_t(r) := \frac{c}{H}\, e^{-(A_t^{(2)}+r)^2/(2H^2)} - \frac{L_4\nu^2}{2}. \tag{20}$$

Lemma 5 (Step-two curvature and noise-averaging shift)**.** *Fix t and $r > 0$ with $\kappa_t(r) > 0$ and $L_3\nu^2 < 2r\kappa_t(r)$. Then*

$$|S_t^{(2)} - S_t^{(2,0)}| \le \frac{L_3\nu^2}{2\kappa_t(r)} < r, \qquad \bar{F}_t^{(2)} \ge \kappa_t(r). \tag{21}$$

In particular $\lambda_2 = \min_t \kappa_t(r)$ is an admissible curvature lower bound in Section 4.

The condition is a small-noise condition: it asks that ν, which is of order σ, be small compared with H after accounting for the signal range. At step one the Gaussian closed form gave a curvature bound for every σ; here the closed form is unavailable, and the Taylor bound only pays off when $L_4\nu^2$ is below the clean curvature.

Proposition 6 (Step-two target shift)**.** *Under the conditions of Lemma 5, deterministically,*

$$|\Delta_t^{(2)}| \le \beta_t^{(2)} := \beta^{(1)} + \frac{L_3\nu^2}{2\kappa_t(r)} \qquad \textit{for every } t. \tag{22}$$

Both terms are of order σ^2 for small σ, and neither decreases with n. The first term is inherited: the step-one shift passes through the step-two filter with constant at most one. The second is new and has the same form as the step-one shift bound of Note 1, Proposition 4.5, with σ^2 replaced by ν^2 and the Gaussian curvature bound replaced by $\kappa_t(r)$.

6 Biased signal error at step two

Corollary 7 (Combined step-two bounds)**.** *Under the conditions of Propositions 4 and 6, with probability at least $1 - 2e^{-x}$, simultaneously for all t,*

$$|\widetilde{S}_t^{(2)} - S_t^{(2)}| \le \mathcal{E}_x^{(2)} + \mathcal{R}_x^{(1)}, \tag{23}$$

$$|\widetilde{S}_t^{(2)} - X^{\mathrm{b}}{}_t^{(2)}| \le \beta_t^{(2)} + \mathcal{E}_x^{(2)} + \mathcal{R}_x^{(1)}. \tag{24}$$

Rates. Suppose the kernel-mass conditions of Note 1 hold at both bandwidths with the same constant C_K, so $N_j \ge nh_j/C_K$, and suppose $d_2 \le 1/2$. Then

$$\mathcal{E}_x^{(2)} \le \frac{3\sigma L_2\Gamma_1}{\lambda_2}\sqrt{\frac{2C_K z_x}{nh_2}}, \qquad \mathcal{R}_x^{(1)} \le \left\{\frac{9L_3A_1^2}{\lambda_1^3} + \frac{2A_1A_2}{\lambda_1^2}\right\}\frac{C_K z_x^{(1)}}{nh_1}, \tag{25}$$

with A_1, A_2 the step-one rate constants of Note 1. The step-two sampling error therefore has the one-step order $\sqrt{z_x/(nh_2)}$, multiplied by Γ_1, plus a carried-over term of the smaller order $z_x/(nh_1)$. Since $h_2 = h_1/a$, the leading term is larger than the step-one sampling error by the factor $\sqrt{a}\,\Gamma_1\lambda_1/\lambda_2$ up to the Bernstein linear term. The target shift is of order σ^2 and does not decrease.

What this says about contraction. Nothing contracts from step one to step two. The sampling error keeps its rate in the local sample size and is amplified, not reduced, by the subtraction. The target shift accumulates. This matches the linear audit in the repository, where the step-k component error has standard deviation of order $(nh_k)^{-1/2}$ with a stage-dependent constant. The constant Γ_1 here is an operator-norm bound and is therefore conservative; the linear audit shows that the exact constant can *decrease* after the first stage because later components are band-pass. Recovering that cancellation would require tracking the Fourier structure of M, not just its norm.

7 Gaps

Gap (Curvature bound for large noise)**.** Lemma 5 needs $L_4\nu^2$ below the clean curvature, roughly $\sigma \lesssim H$. Step one has no such restriction. A truncation argument, bounding $\mathbb{E}\rho''_H(r+\bar{\varepsilon})$ below by ρ''_H at $|r|+\tau$ times $\mathbf{P}(|\bar{\varepsilon}| \le \tau)$, gives a positive bound for every σ but needs a tail bound for $\bar{\varepsilon}^{(2)}$ beyond the second moment, and a separate argument that $S_t^{(2)}$ stays near the active range. Both are available in principle from Lemma 2 applied to $\bar{\varepsilon}_u^{(2)}$, which is itself Lipschitz in ε with constant Γ_1, but are not written out.

Gap (Sharpness of the amplification constant)**.** Γ_1 bounds $\|I-M\|_{\mathrm{op}}$ through row and column sums. It is at least 2 and ignores the cancellation between ε_u and its local average. For the linear filter on a periodic grid the exact factor is the Fourier multiplier $1-w_1(\omega)$, which is small at low frequencies where the step-two kernel concentrates its energy. A sharper constant would need a spectral description of the robust equivalent kernel, which is data dependent.

Gap (Choice of population target)**.** The population target $S_t^{(2)}$ is defined through the surrogate residual, which drops the step-one remainder. The alternative, minimizing $\sum_u \mathbb{E}\rho_H(Y_u^{(2)}-z)a_{u,t}^{(2)}$ with the observed residual, differs from $S_t^{(2)}$ by at most $\mathbb{E}\max_u|R_u^{(1)}|$ after an L^1 version of Lemma 1, which is of order $\mathcal{R}_x^{(1)}$ plus a tail contribution. In simulations, the surrogate target is the one to compute: the clean step-two filter applied to $X-S^{(1)}$, with the loss averaged over the law of $\bar{\varepsilon}^{(2)}$.

Gap (From two steps to k)**.** The argument iterates formally. At step k, the surrogate noise is $\bar{\varepsilon}^{(k)} = (I-M^{(k-1)})\cdots(I-M^{(1)})\varepsilon$ in the differential sense, so the Lipschitz constant becomes $\Gamma_{k-1} \le \prod_{j<k}(1+\gamma_j\sqrt{a_j})$, the carried remainder becomes $\sum_{j<k}\mathcal{R}_x^{(j)}$, and the shift becomes $\sum_{j\le k}$ of noise-averaging terms. Since $\gamma_j \ge 1$, the operator-norm product grows geometrically in k, so this route gives no uniformity in k. The linear audit indicates the true constants are bounded, which again points to the spectral structure of the equivalent kernels. Also, ν_k grows with Γ_{k-1}, which tightens the small-noise condition at each step.

Gap (Inherited from Note 1)**.** Uniformity over the continuum and adaptive choices of h or H remain open. The oracle correction at step two uses the unknown step-two target, curvature, and noise law.

A Proofs

Proof of Lemma 1. Since $\rho''_H > 0$, the objective is strictly convex and $m(y)$ is the unique root of $g(z,y) := \sum_u \rho'_H(y_u-z)a_u$. As $\partial_z g = -\sum_u \rho''_H(y_u-z)a_u < 0$, the implicit function

theorem gives $m \in C^1$ with

$$\frac{\partial m}{\partial y_u} = -\frac{\partial_{y_u} g}{\partial_z g} = \frac{\rho_H''(y_u - m)a_u}{\sum_v \rho_H''(y_v - m)a_v}.$$

These are nonnegative and sum to one. Along the segment $y(s) = y + s(y' - y)$, $s \in [0,1]$,

$$\frac{d}{ds} m(y(s)) = \sum_u \frac{\partial m}{\partial y_u}(y(s))\,(y'_u - y_u) \in \Big[-\max_{u: a_u > 0} |y'_u - y_u|,\ \max_{u: a_u > 0} |y'_u - y_u| \Big],$$

and integrating over s proves (11). □

Proof of Lemma 3. Fix t. The score functional $G_t(\varepsilon) := \sum_u \rho_H'(\bar{Y}_u^{(2)} - S_t^{(2)})\, a_{u,t}^{(2)}$ is smooth in ε, and $\bar{\nabla}_t^{(2)} = G_t - \mathbb{E}G_t$. By the chain rule and (4),

$$\frac{\partial G_t}{\partial \varepsilon_v} = \sum_u a_{u,t}^{(2)}\, \rho_H''(\bar{Y}_u^{(2)} - S_t^{(2)})\,(\delta_{uv} - M_{u,v}) = \big[c^\top (I - M)\big]_v, \qquad 0 \le c_u \le L_2 a_{u,t}^{(2)}.$$

Hence $\|\nabla G_t\|_2 \le \|c\|_2 \|I - M\|_{\mathrm{op}}$. Now $\|c\|_2 \le L_2 \big(\sum_u (a_{u,t}^{(2)})^2\big)^{1/2} \le L_2 N_2^{-1/2}$, and, since $0 \le M_{u,v} \le \gamma_1 a_{v,u}^{(1)}$, the row sums of M are at most γ_1 and the column sums at most $\gamma_1 \bar{a}_1$. By the Schur bound $\|M\|_{\mathrm{op}} \le \sqrt{\|M\|_1 \|M\|_\infty} \le \gamma_1 \sqrt{\bar{a}_1}$, so $\|I - M\|_{\mathrm{op}} \le \Gamma_1$. Lemma 2 with $L = L_2 \Gamma_1 N_2^{-1/2}$ and $s = \sigma L \sqrt{2z}$ gives $\mathbf{P}(|\bar{\nabla}_t^{(2)}| > \sigma L_2 \Gamma_1 \sqrt{2z/N_2}) \le 2e^{-z}$.

The curvature functional $H_t(\varepsilon) := \sum_u \rho_H''(\bar{Y}_u^{(2)} - S_t^{(2)})\, a_{u,t}^{(2)}$ has $\mathbb{E}H_t = \bar{F}_t^{(2)}$ by (7), and the same computation with ρ_H''' in place of ρ_H'' gives $\|\nabla H_t\|_2 \le L_3 \Gamma_1 N_2^{-1/2}$, hence $\mathbf{P}(|H_t - \bar{F}_t^{(2)}| > \sigma L_3 \Gamma_1 \sqrt{2z/N_2}) \le 2e^{-z}$. Take $z = z_x$ and the union over the two functionals and the m evaluation points; the exceptional probability is at most $4me^{-z_x} = e^{-x}$.

For the deterministic bounds, $|\rho_H'''| \le L_3$ and the step-two weights sum to one give the empirical third-derivative bound with $C_3 = L_3/\lambda_2$ in curvature units. Differentiation under the expectation in (7) is justified by the bounded derivatives, so $|\phi_u'''| \le L_3$ and the population-curvature Lipschitz bound of Note 1, Lemma 4.1(iii), follows by the mean value theorem exactly as there. □

Proof of Proposition 4. The population objective $\sum_u \phi_u(\bar{X}_u^{(2)} - z) a_{u,t}^{(2)}$ is finite and coercive by the Lipschitz bound $|\rho_H'| \le 1$ and $\mathbb{E}|\bar{Y}_u^{(2)}| < \infty$, convex, and has curvature $\bar{F}_t^{(2)} \ge \lambda_2 > 0$ at its minimizer, which is therefore unique. The deterministic part of the proof of Note 1, Proposition 4.2, uses only the three bounds of Note 1, Lemma 4.1, on a specified event. On $\Omega_2(x)$ these hold with $C_1 = s_1^{(2)}/\sqrt{\lambda_2}$, $C_2 = s_2^{(2)}/\sqrt{\lambda_2}$, $C_3 = L_3/\lambda_2$, $D_t \ge \sqrt{\lambda_2}$, and $\delta \le d_2$, by Lemma 3. Condition (15) is the corresponding smallness condition, and substitution into the localization and remainder bounds of Note 1 gives (16), exactly as in the proof of Note 1, Corollary 4.3.

For (17), by (5) the observed and surrogate residuals differ by $R_u^{(1)}$. Lemma 1 applied to the step-two fit gives $|\widetilde{S}_t^{(2)} - \bar{S}_t^{(2)}| \le \max_u |R_u^{(1)}| \le \mathcal{R}_x^{(1)}$ on $\Omega_1(x)$. Combine with (10). □

Proof of (19). Centering is immediate. For the second moment, Minkowski's inequality gives $\big(\mathbb{E}(\bar{\varepsilon}_u^{(2)})^2\big)^{1/2} \le \sigma + (\bar{F}_u^{(1)})^{-1} \big(\mathbb{E}(\bar{\nabla}_u^{(1)})^2\big)^{1/2}$, where $\bar{\nabla}_u^{(1)}$ is the normalized step-one score. Its summands are independent and centered, with variances at most $\sigma^2 L_2^2 (a_{v,u}^{(1)})^2$ by the Lipschitz variance bound in the proof of Note 1, Corollary 4.3. Hence $\mathbb{E}(\bar{\nabla}_u^{(1)})^2 \le \sigma^2 L_2^2 / N_1$, and $\bar{F}_u^{(1)} \ge \lambda_1$ gives the claim with $\nu = \sigma(1 + \gamma_1 N_1^{-1/2})$. □

Proof of Lemma 5. Write $\psi = \rho'_H$ and let $\bar{G}(z) := \sum_u a^{(2)}_{u,t} \mathbb{E}\psi(\bar{X}^{(2)}_u + \bar{\varepsilon}^{(2)}_u - z)$ be the averaged score, so that $\bar{G}(S^{(2)}_t) = 0$ and $-\bar{G}' = \sum_u a^{(2)}_{u,t} \mathbb{E}\rho''_H(\cdot)$. Taylor's theorem with $|\psi''| \le L_3$ and $|\rho''''_H| \le L_4$, followed by expectation with $\mathbb{E}\bar{\varepsilon}^{(2)}_u = 0$ and $\mathbb{E}(\bar{\varepsilon}^{(2)}_u)^2 \le \nu^2$, gives for every $r_0 \in \mathbf{R}$

$$|\mathbb{E}\psi(r_0 + \bar{\varepsilon}^{(2)}_u) - \psi(r_0)| \le \frac{L_3\nu^2}{2}, \qquad \mathbb{E}\rho''_H(r_0 + \bar{\varepsilon}^{(2)}_u) \ge \rho''_H(r_0) - \frac{L_4\nu^2}{2}.$$

Let $m := S^{(2,0)}_t$. Since the clean score vanishes at m and the weights sum to one, the first bound gives $|\bar{G}(m)| \le L_3\nu^2/2$. For $|z - m| \le r$ and active u, $|\bar{X}^{(2)}_u - z| \le A^{(2)}_t + r$, because m lies in the active range; the second bound then gives $-\bar{G}'(z) \ge \kappa_t(r)$. Hence $\bar{G}(m + r) \le L_3\nu^2/2 - r\kappa_t(r) < 0$ and $\bar{G}(m - r) \ge -L_3\nu^2/2 + r\kappa_t(r) > 0$. The strictly decreasing function $\bar{G}$ therefore has its unique zero $S^{(2)}_t$ in $(m - r, m + r)$, and the mean value theorem on the interval between m and $S^{(2)}_t$ gives $|S^{(2)}_t - m| \le |\bar{G}(m)|/\kappa_t(r) \le L_3\nu^2/(2\kappa_t(r))$. Finally $\bar{F}^{(2)}_t = -\bar{G}'(S^{(2)}_t) \ge \kappa_t(r)$. □

Proof of Proposition 6. Use the split (18). The first difference is bounded by Lemma 5. For the second, $S^{(2,0)}_t$ and $X^{\mathrm{b}(2)}_t$ are the same clean fit applied to $\bar{X}^{(2)} = X^{(2)} - \Delta^{(1)}$ and to $X^{(2)}$, and Lemma 1 with (1) bounds their difference by $\max_u |\Delta^{(1)}_u| \le \beta^{(1)}$. □

Proof of Corollary 7 and the rates. Add the deterministic bound of Proposition 6 to (17) through the identity (9); the failure probability is unchanged. For the rates, $N_2 \ge nh_2/C_K$ and $d_2 \le 1/2$ give $\mathcal{E}^{(2)}_x \le 3s^{(2)}_1/\lambda_2$ with $s^{(2)}_1 \le \sigma L_2 \Gamma_1 \sqrt{2C_K z_x/(nh_2)}$. The bound on $\mathcal{R}^{(1)}_x$ is Note 1, Corollary 4.4, with the step-one evaluation set equal to the design. □